



Student Performance Dashboard Analytics Project

AI & Data Engineering Graduation Project — **DEPI Program** | Group 4

AI & DATA ENGINEERING

GRADUATION PROJECT



Meet the Team

Group 20 is composed of five AI & Data Engineering students, each bringing a distinct specialization to the project — from pipeline engineering and data cleaning to database architecture, advanced analytics, and quality assurance.

Eman Fadel Kamal Aldeen

Data Pipeline Engineering & Overview
Dashboard

Dina Fouad Esmaiel

Data Cleaning (Python Scripts) &
Student Distribution Analysis

Al Shaimaa Fouad Mohamed

DB Architecture (SQL Schema Design) &
Performance Profiling

Hend Mohammed Hasanein

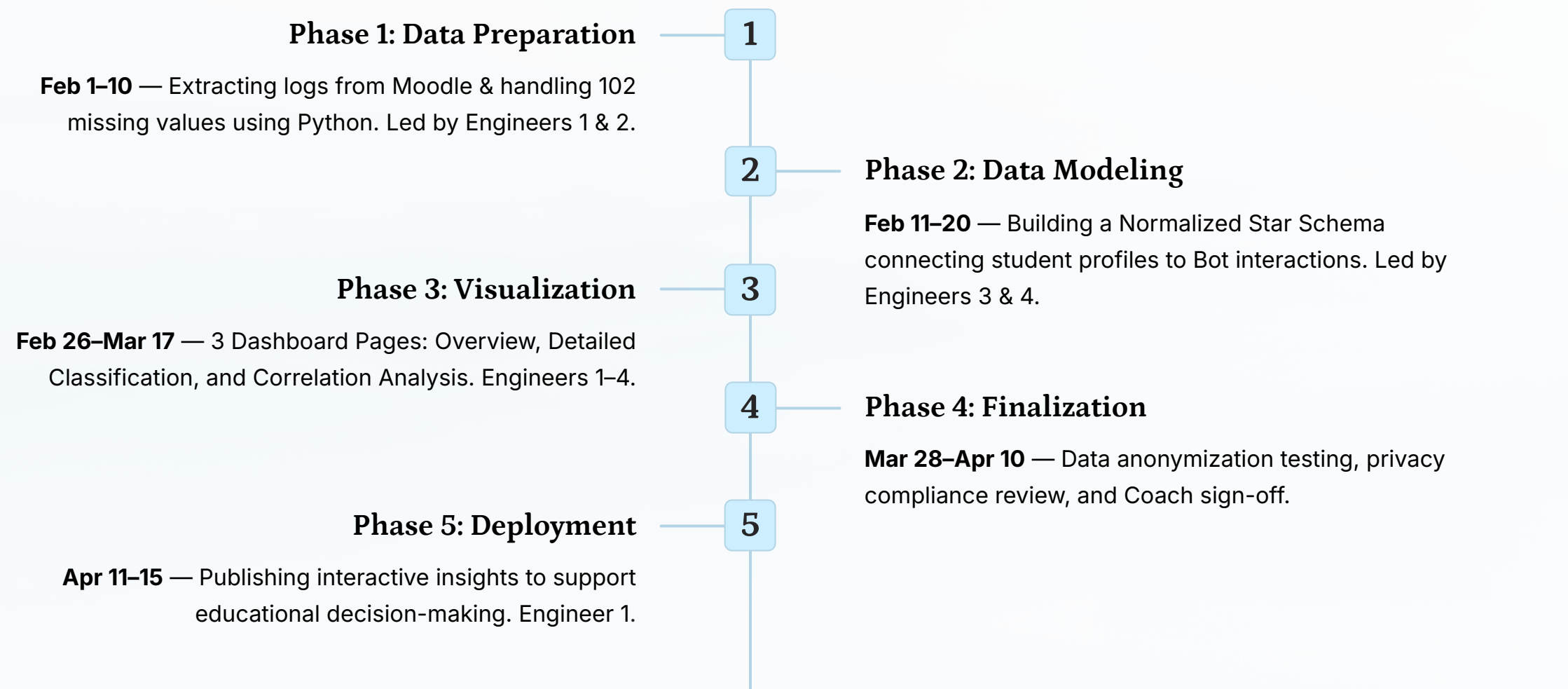
Analytics, Advanced DAX, & Correlation Trends

Eslam Mohammed Hussien

QA, Risk Management, & Data Anonymization Backup

Project Overview & Timeline

The **Student Performance Analytics Dashboard** is a full-cycle data engineering project carried out by 5 AI & Data Engineers, spanning from **February 1 to April 15, 2026**. The project transforms raw Moodle interaction logs into a clean, modeled, and visualized analytics platform that supports evidence-based educational decision-making at Fayoum University.



Responsibilities & Risk Mitigation

Engineer Assignments

Eng 1 — Eman

Data Pipeline Engineering
+ Overview Dashboard
Page

Eng 2 — Dina

Data Cleaning (Python
Scripts) + Student
Distribution Analysis

Eng 3 — Al Shaimaa

DB Architecture (SQL
Schema Design) +
Performance Profiling

Eng 4 — Hend

Analytics, Advanced DAX,
& Correlation Trends

Eng 5 — Eslam

QA, Risk Management, &
Data Anonymization
Backup

A **Weekly Rotating Lead** role coordinates pipeline updates and standardizations across the team.

Risk Mitigation Strategy

→ Backup Support

Engineer 5 assists in Python script fixes if data formatting delays occur post-February 20, ensuring no phase slippage.

→ Pre-built Frameworks

Standard scripts unify mismatched Moodle column headers instantly, eliminating manual reconciliation overhead.

→ Weekly Syncs

Friday technical syncs with the mentor are scheduled to resolve pipeline blockers before they cascade into downstream phases.



Core Toolchain: Python (Pandas / Seaborn) · SQL Server
· Power BI · Git

Presentation Flow

The narrative arc of this project is designed to move evaluators from raw data impact through rigorous analysis to actionable recommendations — each step building credibility for the final strategic conclusions.



Start with Impact

Highlight the massive shift in knowledge gain (+28.03 points) and 100% data cleanliness validated across all 182 students.



Root Causes & Analytics

Show how both groups achieved statistically identical progress using Independent Samples T-Test verification ($P > 0.05$), confirming the integrity of the experimental design.



Student Insights

Present the performance distribution tiers (60.99% Good) and highlight the strong 74% passion-to-gain efficiency conversion rate across the cohort.



The Paradox Trend

Visualize the unexpected and counterintuitive negative correlation ($r = -0.706$) between Total Learning Passion and actual Learning Efficiency — the project's most striking finding.



Recommendations

Conclude with strategic, data-backed actions for standardizing robot-integrated education at Fayoum University, grounded in every analytical finding above.

Structured Methodology

The project follows a rigorous five-phase methodology that maps directly onto the 12 research questions. Each phase has a clear objective, defined steps, and explicit alignment to the analytical questions it answers — ensuring end-to-end traceability from raw data to final recommendations.

1

Data Preparation & Cleaning

Objective: Ensure 100% data completeness for reliable educational modeling.
Exported Moodle logs for 182 students (70 pre/post test items, 32 enjoyment items). Programmatically imputed 102 missing entries via Python. Verified zero missing values across all 243 variables. *Aligns with Q1–Q3.*

2

Data Modeling

Objective: Structure an educational data warehouse optimized for analytics.
Star Schema with `fact_student_performance`, `fact_test_responses`, and dimension tables for students, groups, questions, and assessment items. DAX measures for Knowledge Gain, Enjoyment Lift, and Passion Efficiency ratios. *Aligns with Q4–Q6.*

3

Dashboard Design & Visualization

Objective: Present complex behavioral metrics through intuitive visual interfaces.
Page 1: KPI Cards (28.03 Gain, 82.96 Post-Enjoyment) + Cohort Bars. Page 2: Donut chart for performance tiers. Page 3: Scatter plots for Passion vs. Gain and Enjoyment Lift. *Every visual answers a core question.*

4

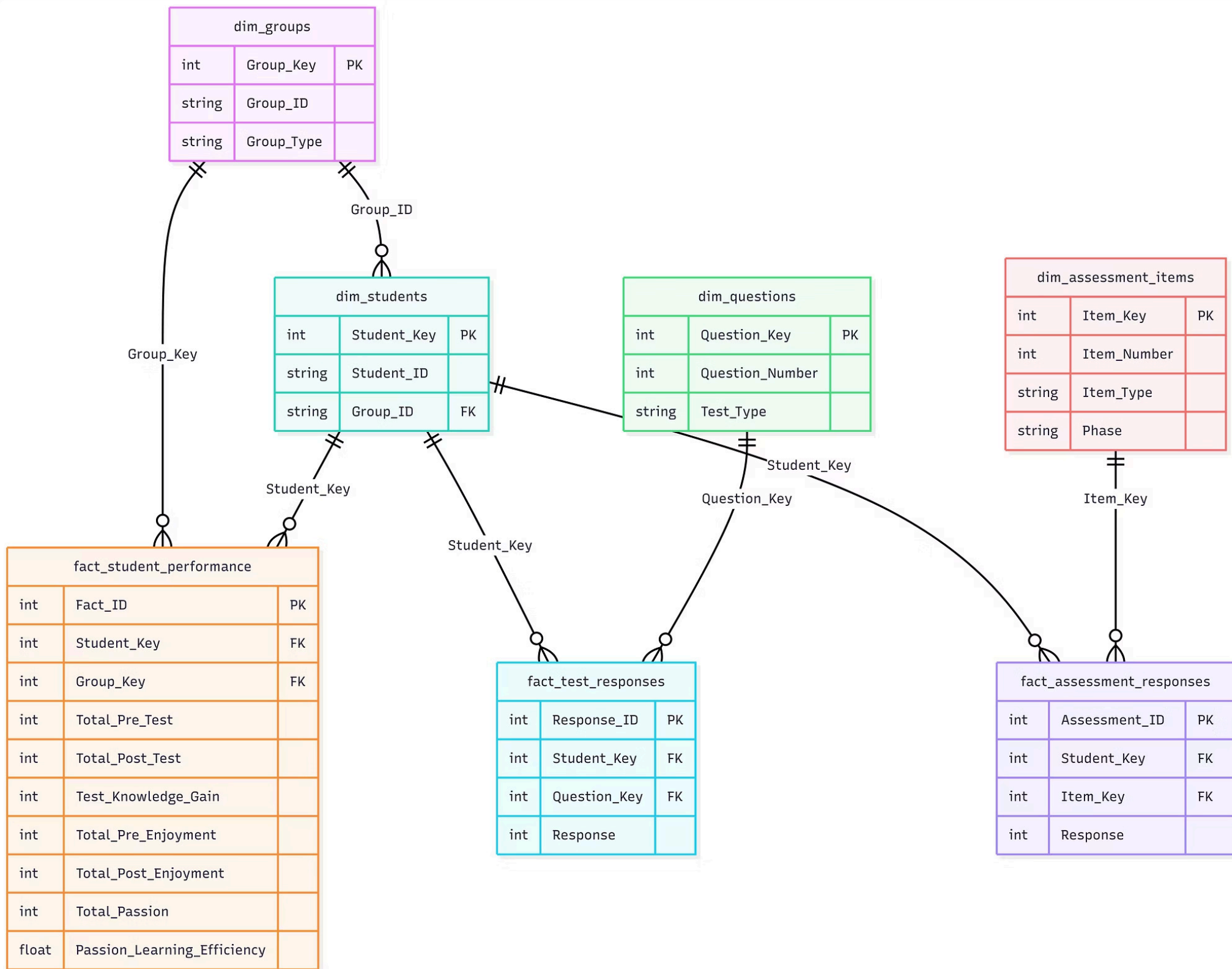
Validation & Testing

Objective: Confirm mathematical and statistical integrity before deployment.
Cross-validated BI dashboard aggregations against raw stats. Executed Independent & Paired T-Tests for group comparisons. Outlier analysis flagged 58 extreme profiles using Z-scores ($Z > 3.0$). *Aligns with Q10–Q12.*

5

Methodology–Question Alignment

Q1 Academic Impact: Pre-to-Post T-Test proves knowledge lift from 27.69 → 55.71.
Q4 Cohort Comparison: Independent Samples T-Test proves GP1/GP2 equivalence.
Q7 Passion Efficiency: DAX ratio reveals 74% passion-to-gain conversion.
Q9 Passion Paradox: Pearson correlation detects $r = -0.706$ between passion and efficiency.



Key Research Questions & Analytical Scope

The project addresses 12 structured research questions organized across four analytical domains. These questions drive every design decision — from the star schema to the DAX measures and dashboard pages — ensuring that every insight is directly tied to an educational outcome.

A. Learning Outcome Analysis

- **Q1:** What is the overall academic impact? — Average knowledge gain reached **+28.03 points**
- **Q2:** How did the intervention affect engagement? — Enjoyment score increased by **+5.86 points**
- **Q3:** What is the effect size? — Cohen's $d \approx 7.16$, far exceeding the 0.8 benchmark

B. Group & Student Performance

- **Q4:** GP1 vs GP2 comparison — GP1 gain: 27.93 vs GP2 gain: **28.11** (near-identical)
- **Q5:** Performance tier distribution — **60.99%** Good, 27.47% Average, 11.54% Excellent
- **Q6:** Outlier detection — **58 flagged anomaly cases** ($Z > 3.0$) for personalized intervention

C. Learning Efficiency

- **Q7:** Passion Learning Efficiency — averaged a strong **74% (0.74)**
- **Q8:** Enjoyment vs. knowledge gain — weak positive link ($r = +0.224$); other factors mediate success
- **Q9:** The Passion-Efficiency Paradox — strong negative correlation ($r = -0.706$) between Total Passion and Efficiency

D. Data-Driven Recommendations

- **Q10:** Standardize robot-assisted setups leveraging proven group equivalence
- **Q11:** Integrate metacognition and self-efficacy variables into Moodle tracking
- **Q12:** Prioritize highly passionate but low-efficiency students with differentiated tasks

Dashboard

Student Performance Analytics Dashboard

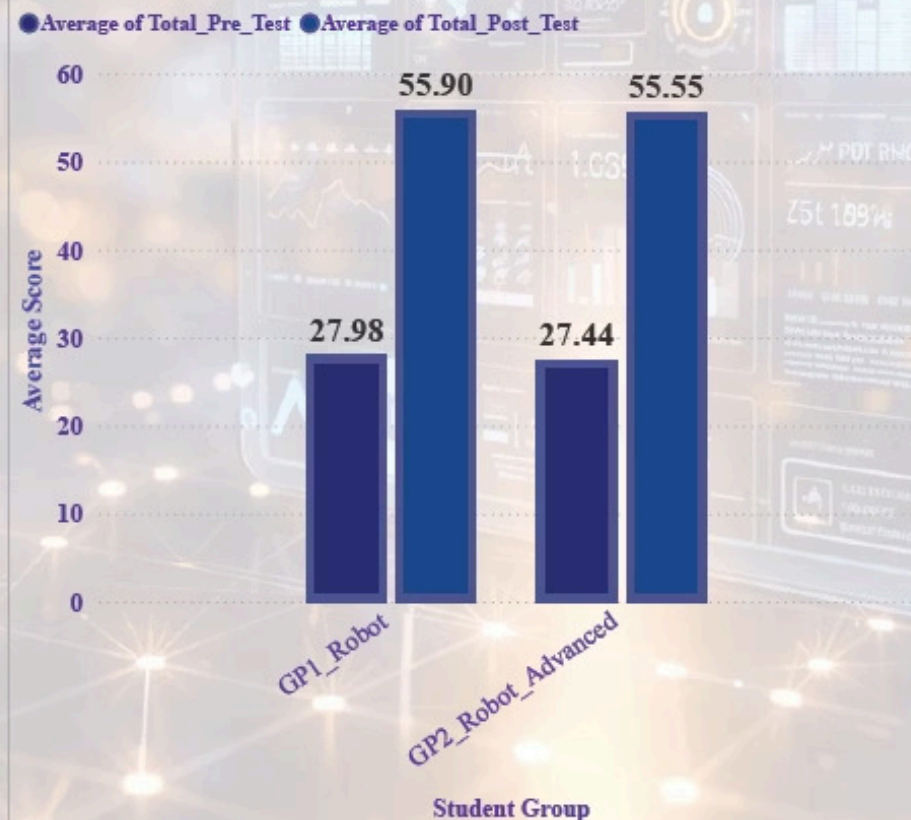
Total Students
182

Learning Efficiency
& Passion
75.55

Pre-Enjoyment Rate
77.10

Post-Enjoyment Rate
82.96

Comparison of Pre-Test and Post-Test Scores by Group



Knowledge Gain Rate
28.03

Avg Pre-Test Score
27.69

Avg Post-Test Score
55.71

Azure Data streaming

The screenshot displays the Microsoft Azure portal interface for a specific SQL database. The browser address bar shows the URL: `portal.azure.com/#@EUI.edu.eg/resource/subscriptions/9c5147c9-5106-48c9-8e6f-b6973803172c/resourceGroups/project-final-server/providers/Micros...`. The page title is "PerformanceDB (project-server-final/PerformanceDB)" and it is identified as an "SQL database".

The left sidebar contains a navigation menu with the following items: Overview (selected), Activity log, Tags, Diagnose and solve problems, Query editor (preview), Mirror database in Fabric (preview), Resource visualizer, Settings, Data management, Integrations, Power Platform, Security, and Auditing.

The main content area is titled "Essentials" and displays various properties of the database in a key-value format:

- Resource group (...): [project-final-server](#)
- Status: Paused
- Location: Sweden Central
- Subscription (move): [Azure for Students](#)
- Subscription ID: 9c5147c9-5106-48c9-8e6f-b6973803172c
- Tags (edit): [Add tags](#)
- Server name: [project-server-final.database.windows.net](#)
- Connection strings: [Show database connection strings](#)
- Pricing tier: [General Purpose - Serverless: Gen5, 1 vCore](#)
- Auto-pause delay: [1 hour](#)
- Earliest restore point: 2026-05-14 00:28 UTC

Below the properties, there is a tabbed interface with the following tabs: Getting started (selected), Monitoring, Properties, Features, Notifications (1), Integrations, and Tutorials.

The "Getting started" tab contains the heading "Start working with your database" and the text "Connect to your database and start working with data with a few simple steps. [Learn more](#)".

At the bottom of the page, there are three icons: a globe, a plug, and a code editor icon.

Hypotheses Investigated

Four formal hypotheses were formulated to provide a rigorous statistical backbone for the project. Each hypothesis is tested using inferential methods and the results directly inform the educational recommendations for Fayoum University's robot-assisted learning program.

H1 — Knowledge Gain from Intervention

H₀: Knowledge gain is ≤ 5 points across the study.

H₁: Knowledge gain is exceptionally large (> 28 points, Cohen's $d \approx 7.16$).

Result: **H₁ confirmed.** The robot-integrated curriculum delivers extraordinary, measurable academic value — far exceeding conventional benchmarks for educational effect size.

H2 — Group Equivalence (GP1 vs GP2)

H₀: Both groups perform equally (mean difference ≈ 0 , $P > 0.05$).

H₁: The advanced group (GP2) vastly outperforms the regular group (GP1).

Result: **H₀ retained.** Groups are statistically equivalent, confirming that the robot-assisted setup works uniformly regardless of cohort type — a critical finding for scalability.

H3 — Enjoyment as a Predictor of Knowledge Gain

H₀: Enjoyment has no statistical relationship with test improvement ($r \leq 0.1$).

H₁: Post-intervention enjoyment acts as a strong predictor of success ($r > 0.5$).

Result: **H₁ partially rejected.** A weak positive link ($r = +0.224$) was detected, suggesting that "fun learning" alone does not drive higher scores — metacognitive and self-efficacy factors must be incorporated into Moodle.

H4 — The Passion-Efficiency Paradox

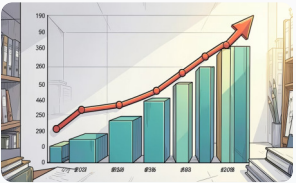
H₀: Passion and learning efficiency are positively or independently linked.

H₁: A paradox exists where extreme passion correlates with lower conversion efficiency ($r = -0.706$).

Result: **H₁ confirmed.** Highly motivated students face structural bottlenecks in Moodle's current design, requiring specialized differentiated tasks to unlock their full learning potential.

🟢 **Key Takeaway:** Three of four hypotheses yield actionable, non-trivial results — making this dataset exceptionally rich for evidence-based curriculum redesign at Fayoum University.

Potential Outcomes & Distinction Criteria



H1 (Academic Gain)

A: Massive gains achieved (+28.03 points).

B: Minor or insignificant gains.

Run Paired t-test. Extremely low P-value ($P < 0.001$) rejects H_0 , proving impact.

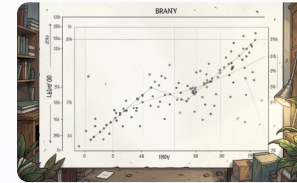


H2 (Group Difference)

A: Group design dictates success.

B: Groups are completely equivalent.

Run Independent t-test. High P-value (0.41 to 0.82) fails to reject H_0 , proving equivalence.

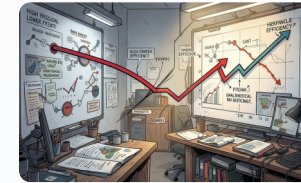


H3 (Enjoyment Link)

A: Enjoyment dictates scores.

B: Weak positive link ($r = 0.224$).

Check Pearson correlation coefficient. $r = 0.224$ supports outcome B; reject strong prediction claim.



H4 (Passion Paradox)

A: High passion causes efficiency loss.

B: Passion improves efficiency.

Scatter plot + Pearson correlation. $r = -0.706$ confirms a powerful negative link (Outcome A).

✅ **Key Takeaway:** Three of four hypotheses yield actionable, non-trivial results — making this dataset exceptionally rich for evidence-based curriculum redesign at Fayoum University.

Business & Educational Implications

Outcome	Action
H1 True (Massive Impact)	Formally adopt and document this robust chatbot framework as a benchmark for national digital course standards (DEPI/DECE).
H2 True (Group Equivalence)	Replicate and scale the baseline design confidently across other university cohorts, saving development costs on variant group versions.
H3 True (Weak Enjoyment Link)	Pivot curriculum design to target hidden mediating variables—such as cognitive self-efficacy and metacognitive tracking—rather than just enjoyment.
H4 True (Passion Paradox)	Optimize Moodle learning paths for top-tier passionate students by introducing advanced, differentiated tasks to resolve the efficiency drop.